

A Search for Anomalous Spectral Energy Distributions in CEERS DR1: An Open-Source JWST Anomaly Detection Pipeline

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JWST NIRC_{am} surveys now provide photometric catalogs large enough to search systematically for galaxies whose SEDs are not well described by standard stellar-population templates. We present an open-source pipeline that fits photo-*z*’s and best-fit template SEDs with EAZY (Brammer et al. 2008), scores per-band residuals with an Isolation Forest + UMAP/DBSCAN ensemble, and cross-matches flagged sources against known contaminants, applied to the CEERS Data Release 1.0 catalog (Cox & et al. 2025).

Of 87,345 CEERS DR1.0 sources ingested, redshift-window ($0.5 \leq z_{\text{phot}} \leq 10$) and coverage cuts retain 68,839 for EAZY fitting. EAZY’s fit-failure sentinel ($\chi^2_{\text{EAZY}} < 0$) flags 2,900 of these (4.2%), excluded from scoring since a non-converged fit yields large artifact residuals, not genuine outliers. The remaining clean sample of 65,939 sources is scored, and the top 2% (1,319 sources) are flagged as anomalous. Cross-matching against Milliquas v8 (Flesch 2023) within 1''5 finds only 5 matches (0.38%): known AGN explain a negligible fraction of the flagged population.

Figure 1 bins the flagged rate in $\Delta z = 0.5$ shells; an OLS fit finds no significant redshift trend ($p = 0.32$). Following the conservative Bayesian framework of Wright et al. (2014) and Lingam & Loeb (2019) for galactic-scale technosignature/biosignature limits, we adopt a prior $P(H_1) = 10^{-8}$ per source and the observed unexplained fraction of the flagged sample as $P(\text{unexplained} | H_0)$; the posterior that an unexplained flagged source reflects non-standard physics is $P(H_1 | \text{unexplained}) \approx 1.12 \times 10^{-8}$, almost unchanged from the prior, since being unexplained is the norm here, not the exception — itself a reproducible upper limit on non-standard astrophysical processes at current CEERS survey depth.

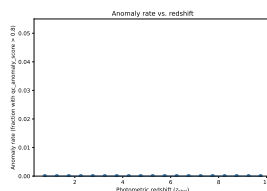


Figure 1. Flagged rate vs. z , clean sample ($N = 65,939$); $p = 0.32$.

Code: <https://github.com/Menrae/jwst-sed-anomaly>.

Software: astropy, eazy-py (Brammer et al. 2008), scikit-learn, umap-learn, xarray

REFERENCES

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